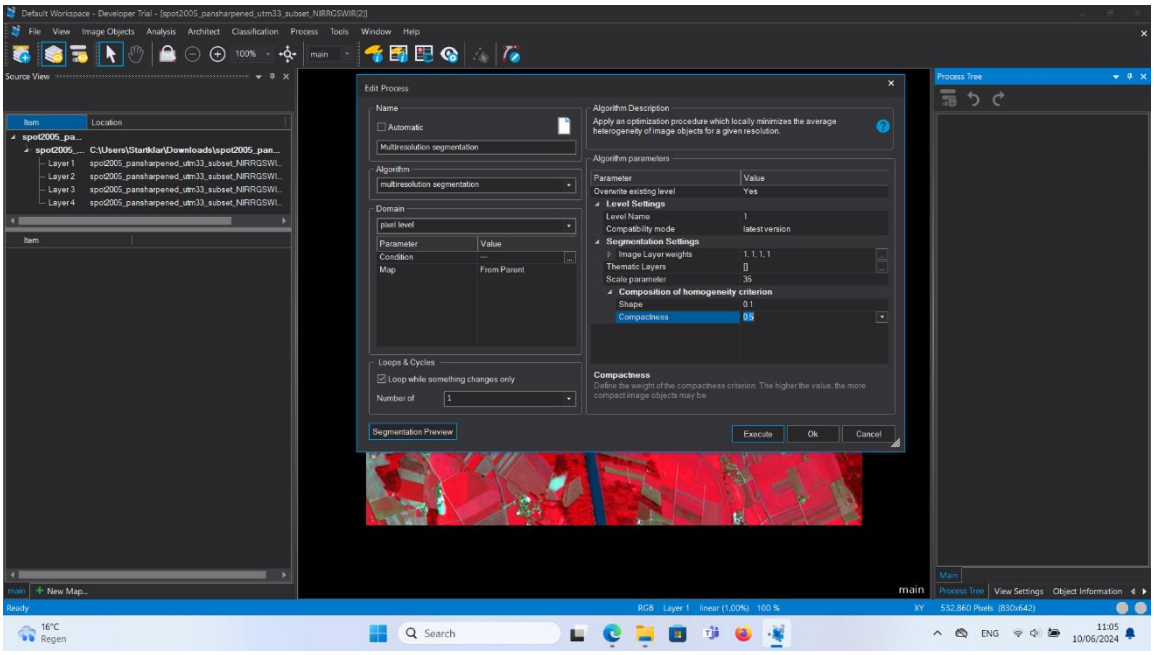
Assignment 1

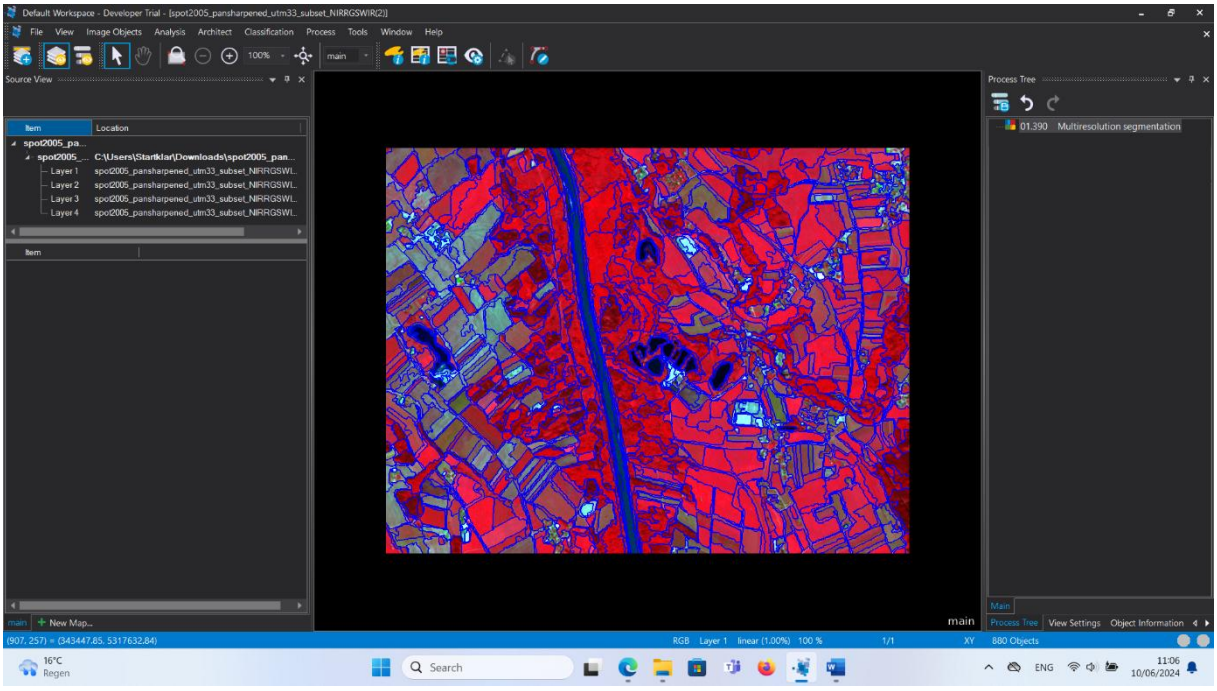
Advanced Remote Sensing

Multi Resolution Segmentation with Supervised Classification

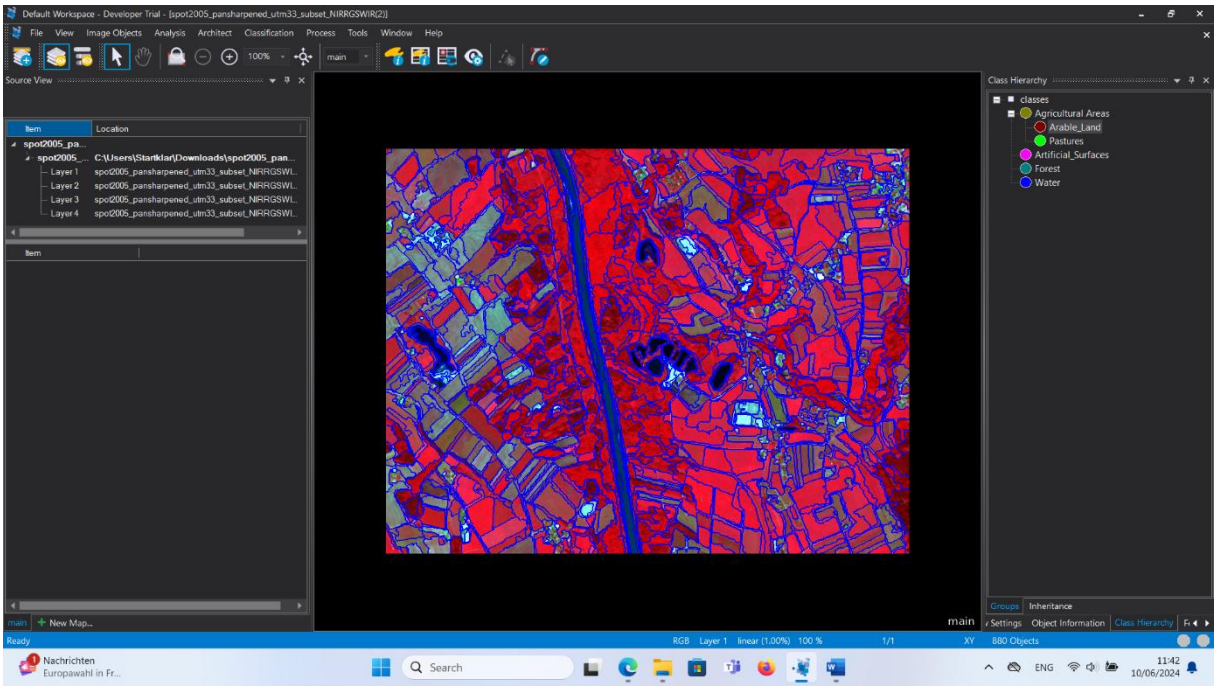
1.Multi Resolution Segmentation



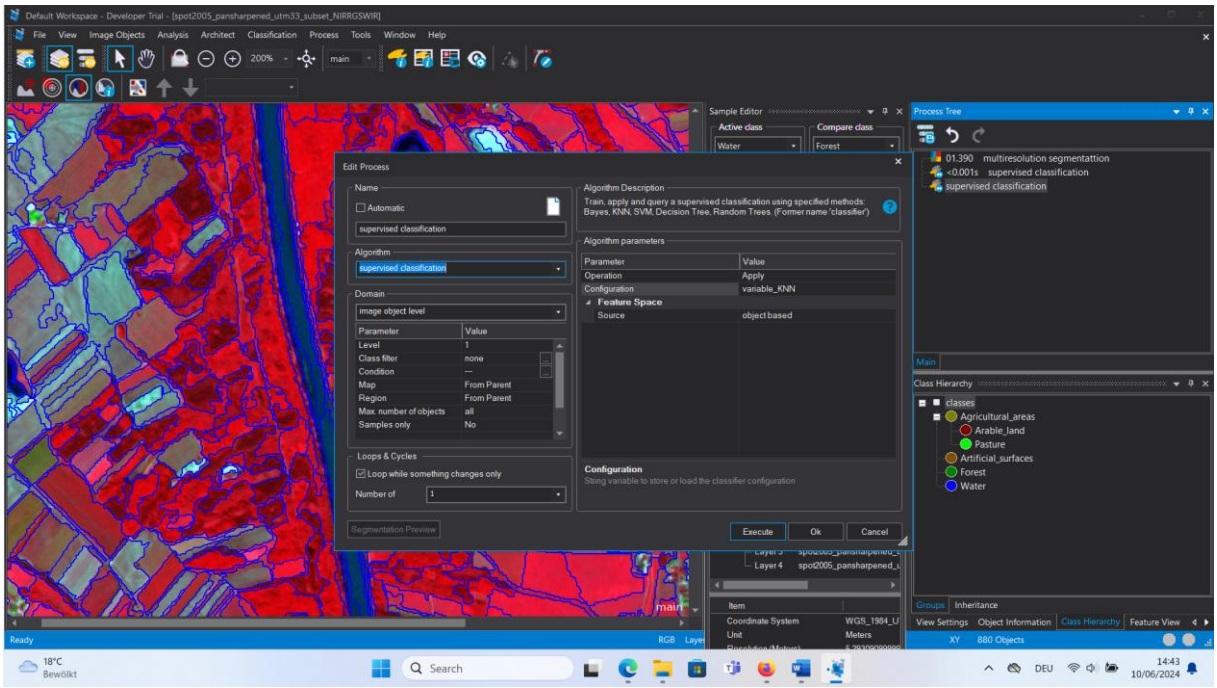
2.



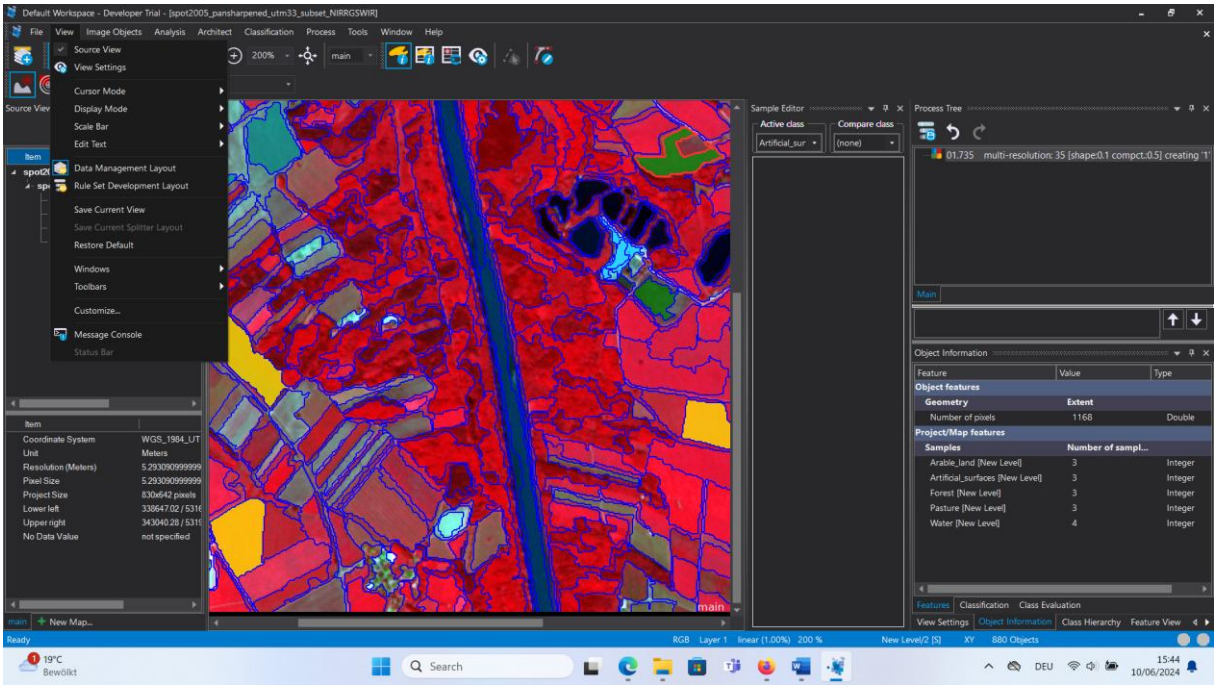
3.


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR(2)]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Item Location
spot2005_pa...
spot2005_... C:\Users\Startklar\Downloads\spot2005_pan...
Layer 1 spot2005_pansharpened_utm33_subset_NIRRGSWI..
Layer 2 spot2005_pansharpened_utm33_subset_NIRRGSWI..
Layer 3 spot2005_pansharpened_utm33_subset_NIRRGSWI..
Layer 4 spot2005_pansharpened_utm33_subset_NIRRGSWI..
Class Hierarchy
classes
Agricultural Areas
Arable_Land
Pastures
Artificial_Surfaces
Forest
Water
main
Groups Inheritance
Settings Object Information Class Hierarchy
Ready
880 Objects
Nachrichten
Europawahl in Fr...
Search
ENG
11:42
10/06/2024

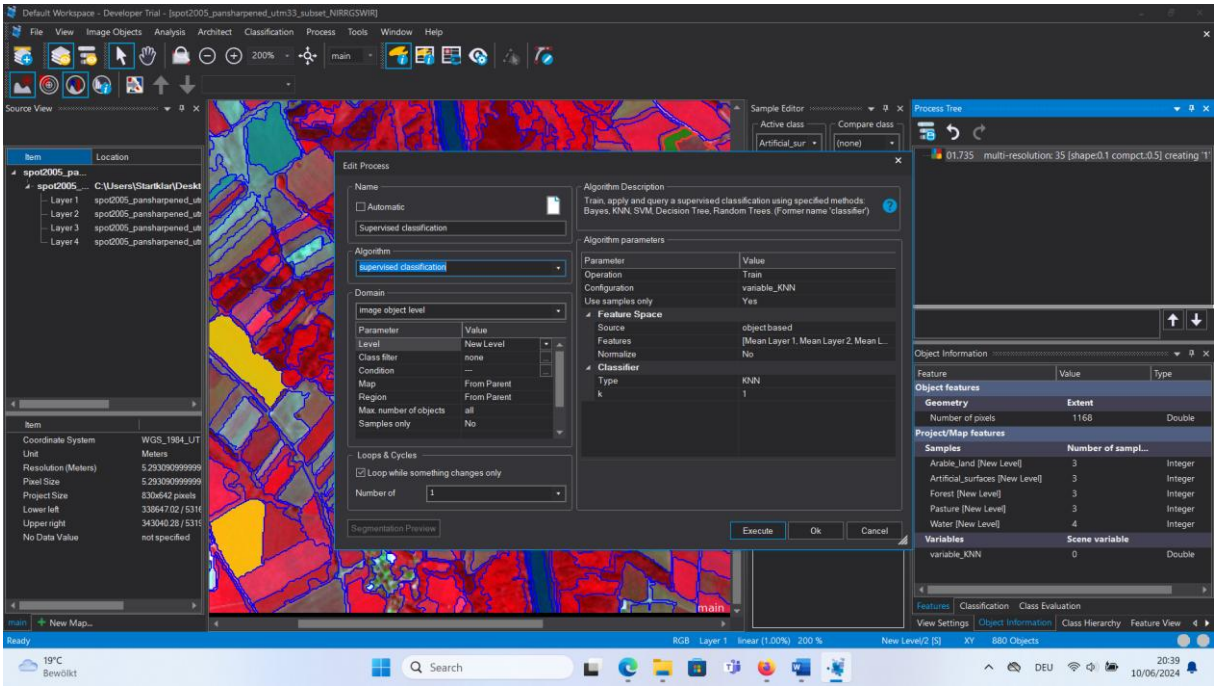
4.


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Sample Editor
Active class
Water
Compare class
Forest
Process Tree
01.390 multiresolution segmentattion
<0.001s supervised classification
supervised classification
Edit Process
Name
Automatic
supervised classification
Algorithm
supervised classification
Domain
image object level
Parameter Value
Level 1
Class filter none
Condition
Map From Parent
Region From Parent
Max. number of objects all
Samples only No
Loops & Cycles
Loop while something changes only
Number of 1
Segmentation Preview
Algorithm Description
Train, apply and query a supervised classification using specified methods: Bayes, KNN, SVM, Decision Tree, Random Trees. (Former name 'classifier')
Algorithm parameters
Parameter Value
Operation Apply
Configuration variable_KNN
Feature Space
Source object based
Configuration
String variable to store or load the classifier configuration
Execute
Ok
Cancel
Class Hierarchy
classes
Agricultural_areas
Arable_land
Pasture
Artificial_surfaces
Forest
Water
Coordinate System WGS_1984_U
Unit Meters
Groups Inheritance
View Settings Object Information Class Hierarchy Feature View
880 Objects
18°C
Bewölkt
Search
DEU
14:43
10/06/2024

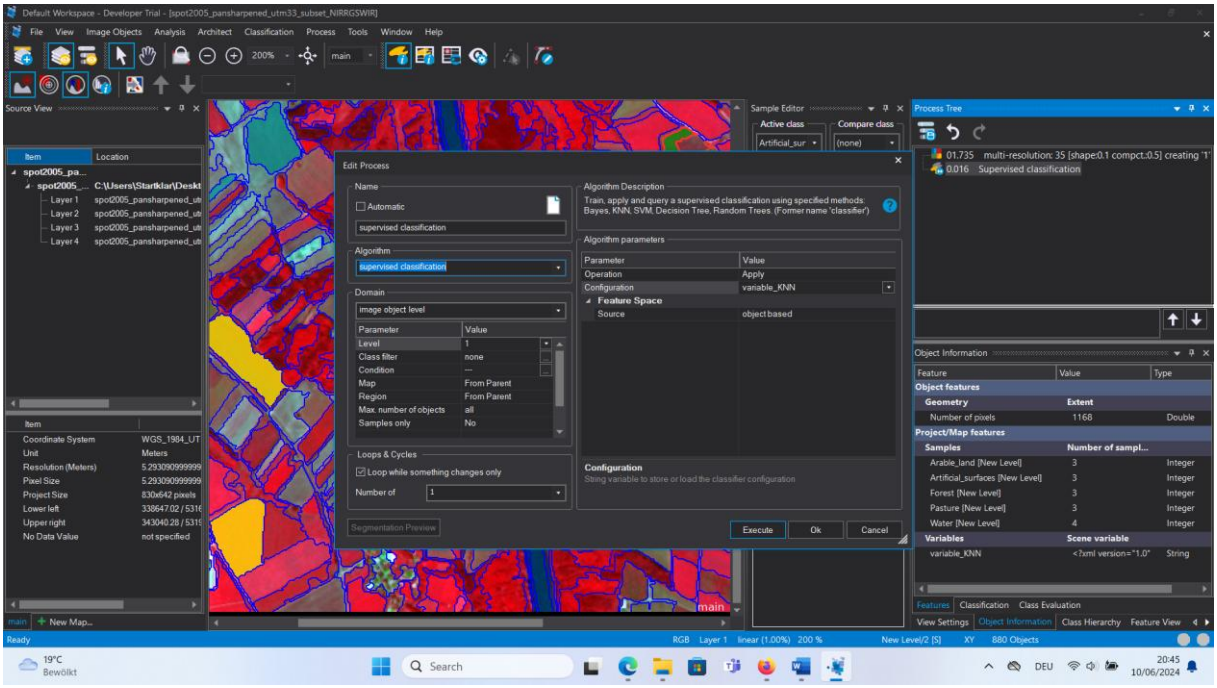
5.


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
View Settings
Cursor Mode
Display Mode
Scale Bar
Edit Text
Data Management Layout
Rule Set Development Layout
Save Current View
Save Current Splitter Layout
Restore Default
Windows
Toolbars
Customize...
Message Console
Status Bar
Sample Editor
Active class
Compare class
Process Tree
Object Information
Feature
Value
Type
Object features
Geometry
Extent
Number of pixels
1168
Double
Project/Map features
Samples
Number of sampl...
Arable_land [New Level]
Artificial_surfaces [New Level]
Forest [New Level]
Pasture [New Level]
Water [New Level]
Integer
Coordinate System
Unit
Resolution (Meters)
Pixel Size
Project Size
Lower left
Upper right
No Data Value
WGS_1984_UT
Meters
830x642 pixels
not specified
Features
Classification
Class Evaluation
View Settings
Object Information
Class Hierarchy
Feature View
Ready
Search
DEU
15:44
10/06/2024

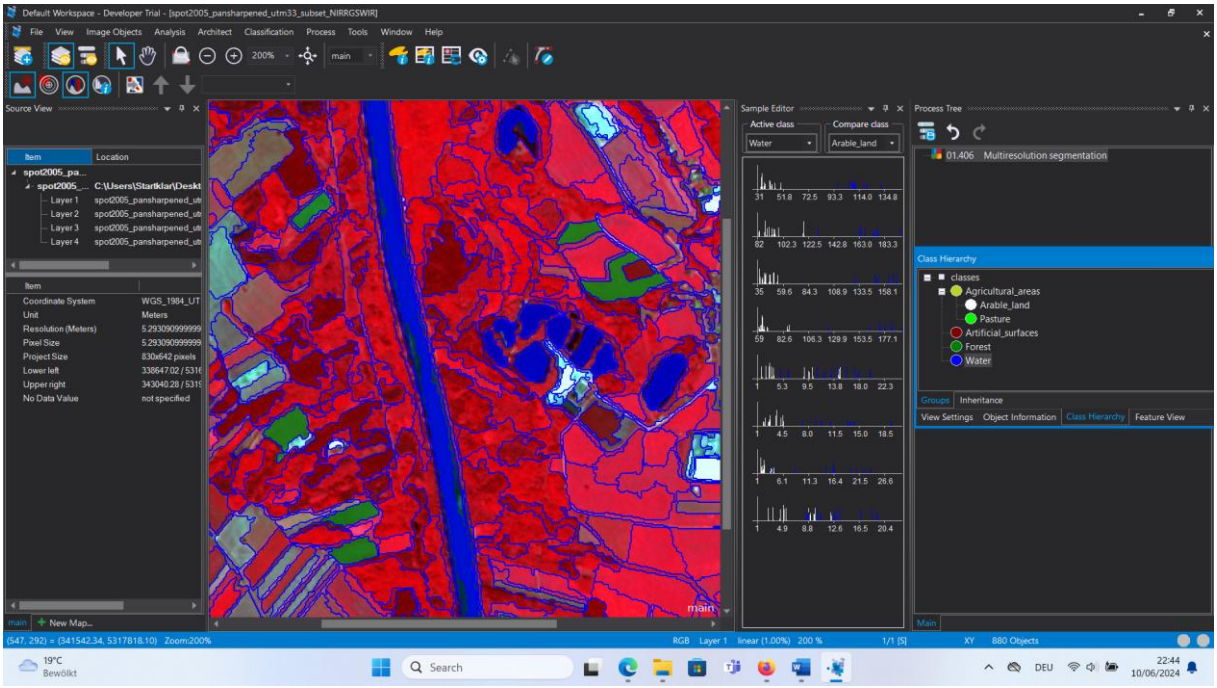
6.


Edit Process
Name
Automatic
Supervised classification
Algorithm
supervised classification
Domain
image object level
Algorithm Description
Train, apply and query a supervised classification using specified methods: Bayes, KNN, SVM, Decision Tree, Random Trees. (Former name 'classifier')
Algorithm parameters
Parameter
Value
Operation
Train
Configuration
variable_KNN
Use samples only
Yes
Feature Space
Source
object based
Features
[Mean Layer 1, Mean Layer 2, Mean L...
Normalize
No
Classifier
Type
KNN
k
1
Loops & Cycles
Loop while something changes only
Number of
Segmentation Preview
Execute
Ok
Cancel
Sample Editor
Process Tree
Object Information
Variables
Scene variable
variable_KNN
Ready
Search
DEU
20:39
10/06/2024

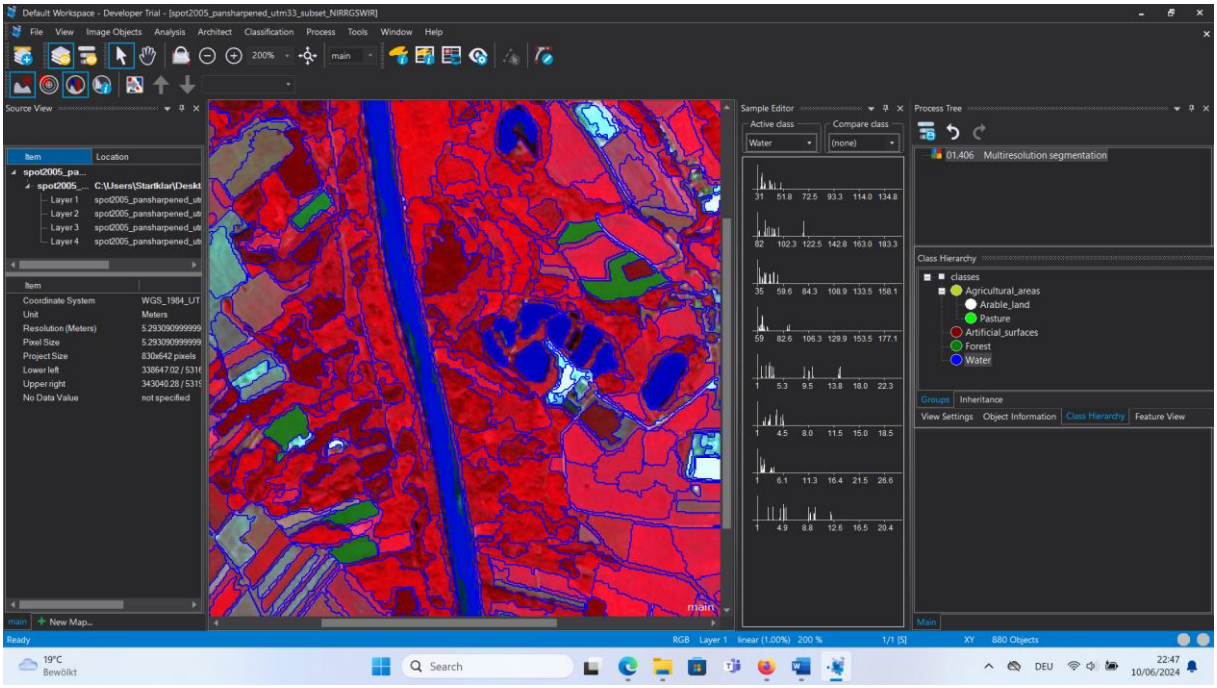
7.


Edit Process
Name
Automatic
supervised classification
Algorithm
supervised classification
Domain
image object level
Loops & Cycles
Loop while something changes only
Number of
Algorithm Description
Train, apply and query a supervised classification using specified methods: Bayes, KNN, SVM, Decision Tree, Random Trees. (Former name 'classifier')
Algorithm parameters
Parameter
Value
Operation
Apply
Configuration
variable_KNN
Feature Space
Source
object based
Level
Class filter
Condition
Map
Region
Max. number of objects
Samples only
From Parent
all
No
Configuration
String variable to store or load the classifier configuration
Segmentation Preview
Execute
Ok
Cancel
Process Tree
01.735 multi-resolution: 35 [shape:0.1 compct.:0.5] creating '1'
0.016 Supervised classification
Object Information
Feature
Value
Type
Object features
Geometry
Extent
Number of pixels
1168
Double
Project/Map features
Samples
Number of sampl...
Arable_land [New Level]
Artificial_surfaces [New Level]
Forest [New Level]
Pasture [New Level]
Water [New Level]
Variables
Scene variable
variable_KNN
<?xml version="1.0"
String
Integer
Sample Editor
Active class
Compare class
Source View
Item
Location
Coordinate System
Unit
Resolution (Meters)
Pixel Size
Project Size
Lower left
Upper right
No Data Value
WGS_1984_UT
Meters
not specified
Ready
Search

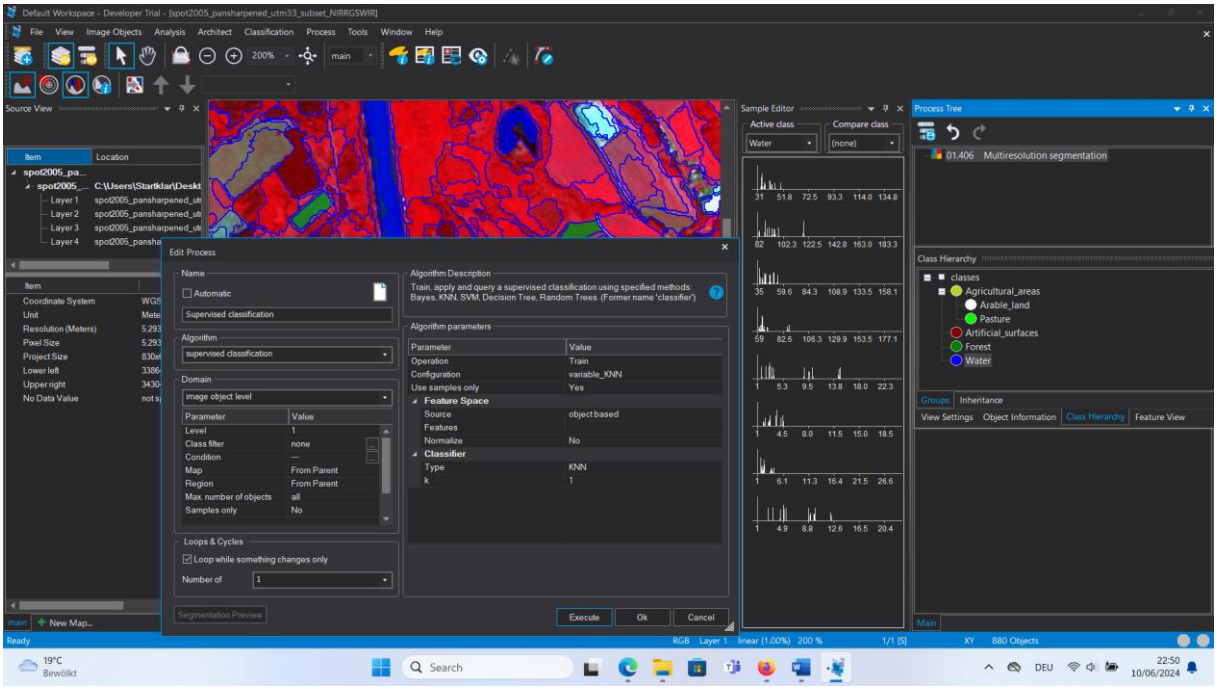
8.


Source View
Sample Editor
Active class
Water
Compare class
Arable_land
Process Tree
01.406 Multiresolution segmentation
Class Hierarchy
classes
Agricultural_areas
Arable_land
Pasture
Artificial_surfaces
Forest
Water
Groups
Inheritance
View Settings
Object Information
Class Hierarchy
Feature View
Item
Location
Coordinate System
Unit
Resolution (Meters)
Pixel Size
Project Size
Lower left
Upper right
No Data Value
WGS_1984_UT
Meters
not specified
Search

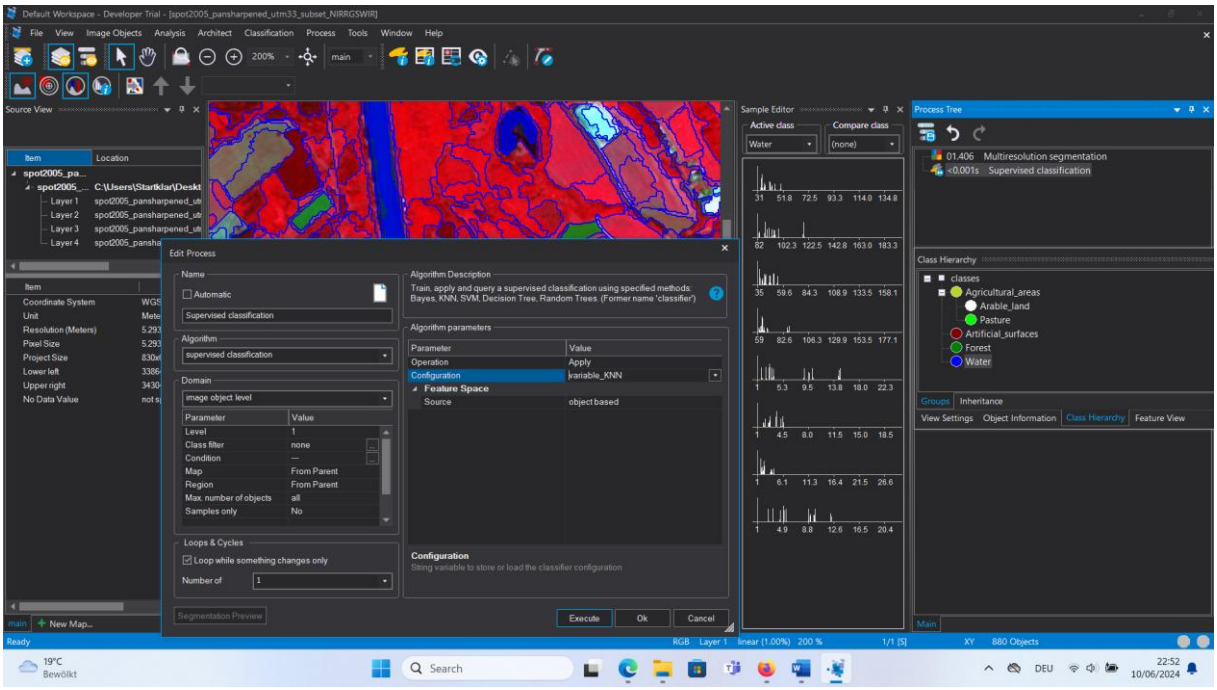
9.


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Sample Editor
Active class
Compare class
Water
(none)
Process Tree
01.406 Multiresolution segmentation
Class Hierarchy
classes
Agricultural_areas
Arable_land
Pasture
Artificial_surfaces
Forest
Water
Groups Inheritance
View Settings Object Information Class Hierarchy Feature View
Ready
Search
DEU
22:47
10/06/2024

10.


Edit Process
Name
Automatic
Supervised classification
Algorithm
supervised classification
Domain
image object level
Algorithm Description
Train, apply and query a supervised classification using specified methods: Bayes, KNN, SVM, Decision Tree, Random Trees. (Former name 'classifier')
Algorithm parameters
Parameter
Value
Operation
Train
Configuration
variable_KNN
Use samples only
Yes
Feature Space
Source
object based
Features
Normalize
No
Classifier
Type
KNN
k
1
Loops & Cycles
Loop while something changes only
Number of
Segmentation Preview
Execute
Ok
Cancel
Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
Source View
Sample Editor
Process Tree
01.406 Multiresolution segmentation
Class Hierarchy
classes
Agricultural_areas
Arable_land
Pasture
Artificial_surfaces
Forest
Water
Ready
Search
DEU
22:50
10/06/2024

11.


Edit Process
Name
Automatic
Supervised classification
Algorithm
supervised classification
Domain
image object level
Parameter
Value
Level
1
Class filter
none
Condition
Map
From Parent
Region
From Parent
Max. number of objects
all
Samples only
No
Loops & Cycles
Loop while something changes only
Number of
Algorithm Description
Train, apply and query a supervised classification using specified methods: Bayes, KNN, SVM, Decision Tree, Random Trees. (Former name 'classifier')
Algorithm parameters
Operation
Apply
Configuration
Feature Space
Source
object based
Configuration
String variable to store or load the classifier configuration
Segmentation Preview
Execute
Ok
Cancel
Sample Editor
Active class
Water
Compare class
(none)
Process Tree
01.406 Multiresolution segmentation
<0.001s Supervised classification
Class Hierarchy
classes
Agricultural_areas
Arable_land
Pasture
Artificial_surfaces
Forest
Water
Groups
Inheritance
View Settings
Object Information
Class Hierarchy
Feature View
Source View
Item
Location
Ready
880 Objects
Search
19°C
Bewölkt
DEU
22:52
10/06/2024